



Introduction to NLP



1. What is NLP?

Imagine you have a **very smart robot friend** 🤖

You tell it:

"Hey, book me a flight tomorrow"

For you, it's simple.

But for a computer, this is just **text**:

"Hey, book me a flight tomorrow"



NLP (Natural Language Processing) is what helps computers:

- **Understand human language**
- **Process it**
- **Respond intelligently**



In simple words:

NLP = Teaching machines to **read, understand, and talk like humans**



2. Real-Life Examples (You already use NLP daily)

- Voice assistants (Google Assistant, Alexa)
 - Chatbots (customer support)
 - Spam detection (Gmail)
 - Recommendations (Netflix, Amazon)
 - Auto-correct & predictive text
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? 3. Why was NLP created? (Problem → Evolution)

● Problem:

Computers only understand:

- Numbers (0s and 1s)
- Structured data (tables, rows)

But humans communicate using:

- Text
- Speech
- Emotions
- Context

👉 Huge gap between **human language** and **machine understanding**

🧑 Old Approach (Rule-Based Systems)

- Humans wrote rules like:
 - “If sentence contains ‘buy’, mark as intent”
 - Problems:
 - Not scalable
 - Breaks with new sentences
 - No understanding
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🧠 New Approach (Modern NLP)

- Use **Machine Learning + Deep Learning**
 - Model learns patterns from data
 - Example:
 - Learns what “booking a flight” looks like from thousands of examples
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🧩 4. Core Concepts of NLP

Let's break NLP into simple building blocks:

1. Text Preprocessing

👉 Cleaning raw text

Example:

"I LOVE AI!!!" → "love ai"

Includes:

- Lowercasing
 - Removing punctuation
 - Removing stopwords (like "is", "the")
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2. Tokenization

👉 Breaking sentence into words

"I love AI" → ["I", "love", "AI"]

3. Feature Extraction (Very Important)

👉 Convert text → numbers (because machines need numbers)

Example:

"AI is cool" → [0.2, 0.8, 0.5]

Techniques:

- Bag of Words (BOW)
 - TF-IDF
 - Word Embeddings (Word2Vec)
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4. NLP Tasks (What we actually do)

- **Text Classification** → Spam vs Not Spam
- **Sentiment Analysis** → Positive / Negative
- **Named Entity Recognition (NER)** → Identify names, places
- **Translation** → English → Hindi

- Chatbots / Q&A systems
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⚙️ 5. How NLP Works (Intuitive → Technical)

💡 Intuitive Flow

Text Input → Clean → Break → Convert to Numbers → Model → Output

Example:

"I love this movie"

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Clean → Tokenize → Convert → Model predicts → Positive sentiment

⚙️ Technical Flow

1. Input text
 2. Preprocessing
 3. Tokenization
 4. Vectorization (BOW / TF-IDF / Embeddings)
 5. Model (ML / DL / Transformers)
 6. Output prediction
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🔥 6. Where You'll Use NLP (Your Context - Data Engineer + AI)

Since you're moving towards **AI + Data Engineering**, NLP is used in:

- Log analysis (detect anomalies in logs)
 - Healthcare text analysis (doctor notes, prescriptions)
 - Chatbots for business workflows
 - Document processing pipelines (PDF → insights)
 - RAG systems (very important for GenAI)
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💡 Final Mental Model

Think of NLP as:

🗣️ "Translator between Human Language and Machine Language"



3. Real-World Use Cases

This exact pipeline is used in:



E-commerce

- Review analysis (Amazon, Flipkart)



Healthcare (Your domain)

- Doctor notes sentiment
- Patient feedback analysis



Customer Support

- Detect angry customers
- Route tickets automatically



Data Engineering Use Case (VERY IMPORTANT FOR YOU)

- Build pipeline:
 - Ingest → Transform → Apply NLP → Store insights
 - Example:
 - S3 → Glue → NLP model → Snowflake
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Final Practical Understanding



NLP pipeline =

Text → Clean → Tokenize → Vectorize → Model → Prediction
